Chulalongkorn University
จุฬาลงกรณ์มหาวิทยาลัย
Pillar of the Kingdom

Location choice and tax responsiveness of foreign multinationals: Evidence from ASEAN countries

PIER Economic Seminar

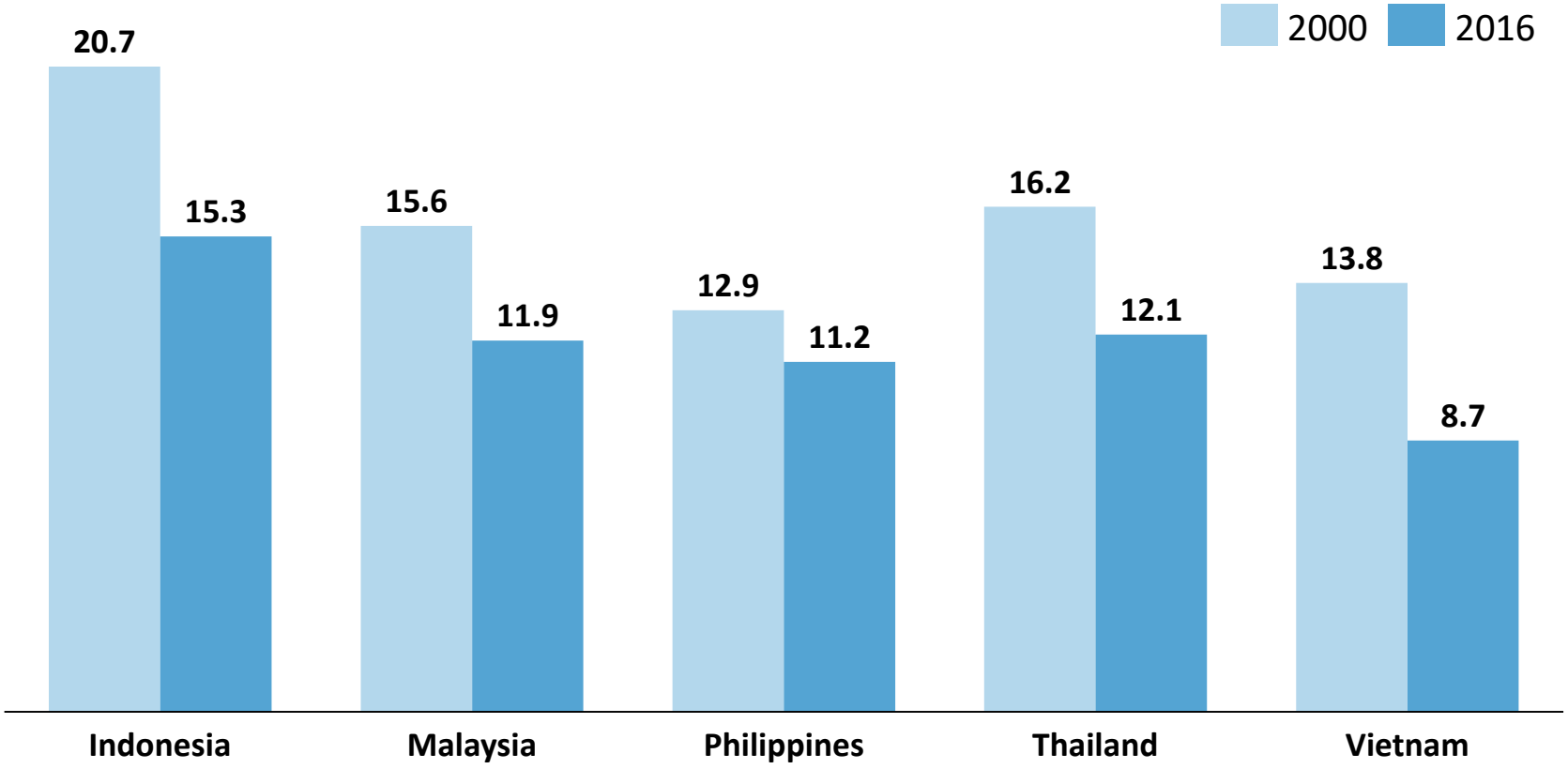
November 2018

Athiphat Muthitacharoen, PhD
Chulalongkorn University
athiphat.m@chula.ac.th

This project was financially supported by Puey Ungphakorn Institute for Economic Research, Bank of Thailand.

Taxes on FDI in ASEAN5 have fallen considerably over the past decades

Bilateral EATR under maximum tax incentives: 2000 and 2016 (%)



Note: The bilateral EATR is averaged across 25 parent countries
Source: Author's estimate

This underlines the importance of understanding tax responsiveness of MNEs

How do MNEs respond to taxes in their investment location choice decision?

Helping address tax competition pressure

Serving as inputs for cost-benefit exercises

Although well studied for advanced economics, empirical evidence for developing countries is very limited

Overview of the Study

Research Questions	Research Design	Key Findings
1 To what extent, do location choice of MNEs respond to taxes in the context of developing countries?	▪ Use data of MNEs which established affiliates in ASEAN5 over 2000-2016	▪ Tax is important but its importance is smaller in the low-tax environment
2 How does the tax responsiveness vary for different MNEs?	▪ Estimate conditional logit model ▪ Compute bilateral effective average tax rate (EATR)	▪ Tax responsiveness is smaller for high-tech firms, return investors and those connected to tax haven.

Note: ASEAN5 = Indonesia, Malaysia, the Philippines, Thailand, and Vietnam

Presentation Outline

1. Introduction

2. Empirical Strategy

3. Findings

Presentation Outline

1. Introduction

2. Empirical Strategy

3. Findings

MNE Data

Sample construction

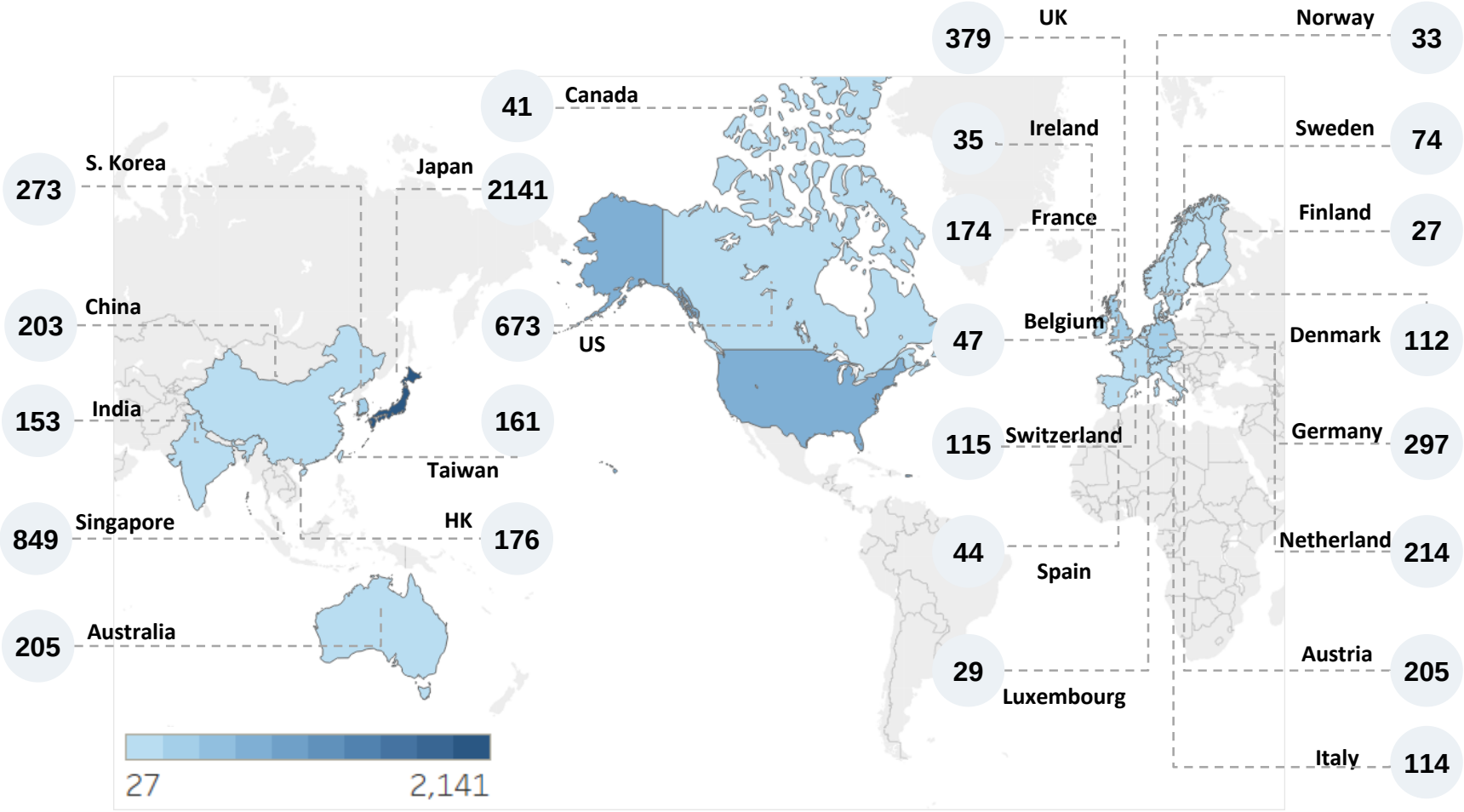
- Main source: Orbis
  - Financial account
  - Corporate group structure
- Focus on location decisions of newly-established foreign subsidiaries in ASEAN5
- **Foreign subsidiary**: At least 50% of the shares ultimately owned by a foreign firm

Baseline sample

- Host countries: ASEAN5 (Indonesia, Malaysia, the Philippines, Thailand, and Vietnam)
- Home countries: Top 25 parent countries
- Time period: 2000-2016
- Baseline sample: 6,616 foreign affiliates

Geographical distribution of parent countries

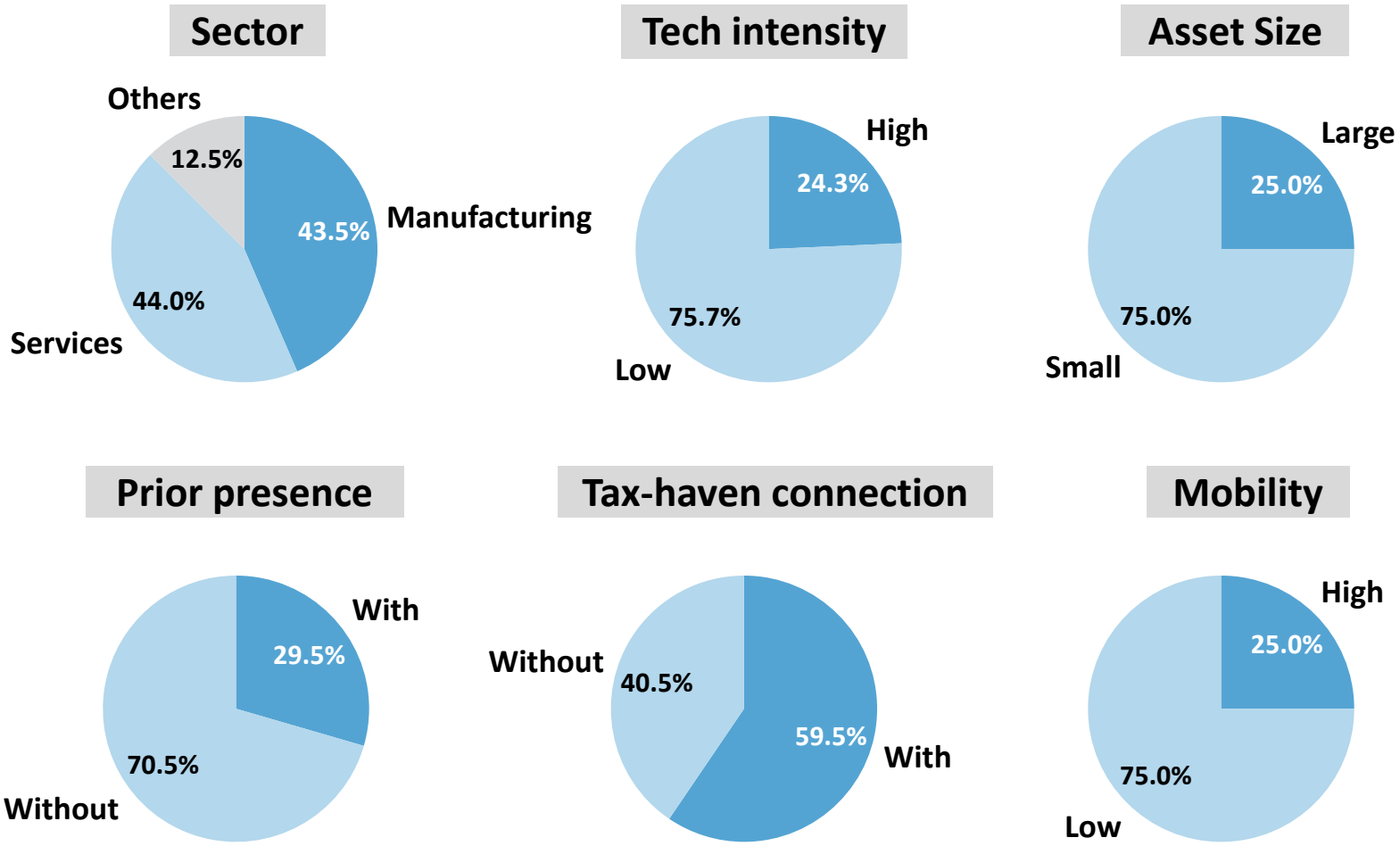
in circle indicates # of ASEAN5 subsidiaries that were established during 2000-2016



Note: Darker shade indicates larger number of subsidiaries

Source: Orbis

Breakdown of firms across dimensions for the heterogeneity analysis



Source: Author's estimates

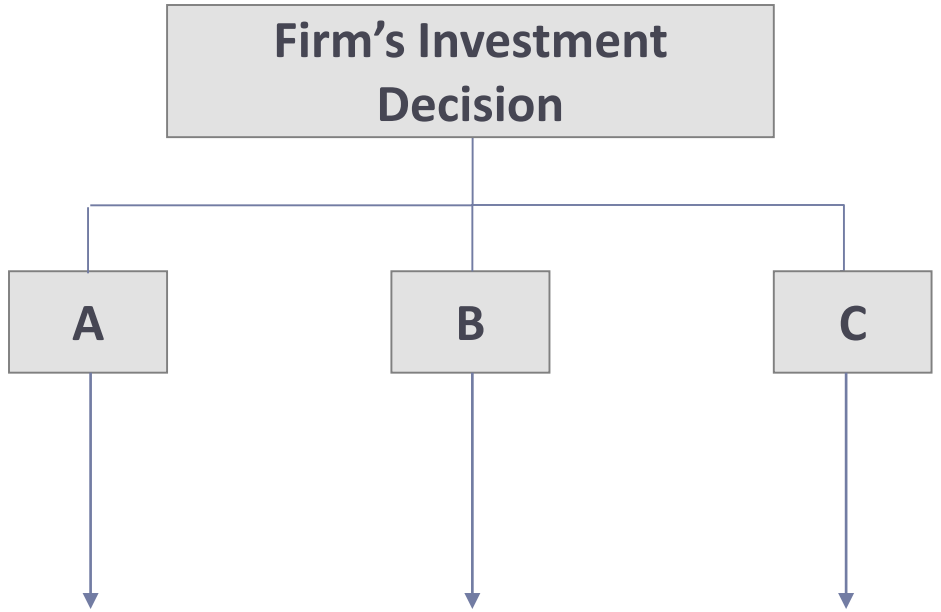
Decision tree on firm's investment decisions

Our Focus

1 **Investment Location Decision**

Relevant tax rate:

Effective Average Tax Rate (EATR)



2 **Investment Scale Decision Conditional on the Chosen Location**

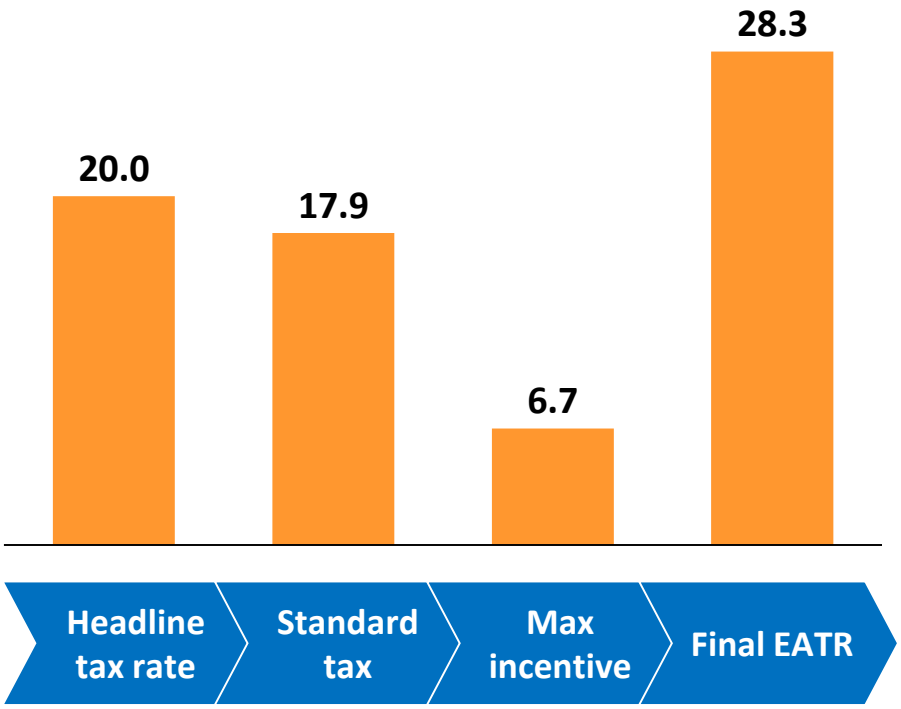
Relevant tax rate: **Effective Marginal Tax Rate (EMTR)**

The EATR construction takes into account both host-country and international tax provisions

Key tax laws incorporated

- Host-country tax holiday
- Withholding tax on repatriated income
- Parent country's treatment of foreign-sourced income
- Relief of double taxation: Unilateral and DTA (Ordinary, Indirect, Tax-sparing credits)

Example: Taxes for an investment project of a US firm in Thailand (Max Incentives: 2016)



Source: Author's estimates

Conceptual framework

- Use Conditional logit model (McFadden, 1974, 1976) to model the location choice as a function of tax and other factors

$$\Pi_{ij} = \beta' X_{ij} + \varepsilon_{ij}$$

where Π_{ij} = profit for firm i if located in country j, X_{ij} = vector of location-specific control variables, and ε_{ij} = error term

- Other control variables: 1) Size of the economy, 2) Geographical proximity, 3) Cultural proximity, 4) Governance quality, and 5) Infrastructure adequacy
- Performing Extreme Bounds Analysis (EBA) to test sensitivity regarding the choice of the control variables

Summary statistics

Summary statistics for variables used in the baseline regression analysis				
Variable	**N**	**Mean**	**Median**	**SD**
Bilateral EATR	33,080	0.157	0.133	0.078
Distance	33,080	6,497	5,329	4,143
Colonial relationship	33,080	0.045	0.000	0.207
Real GDP	33,080	317,700	244,600	227,900
Governance quality	33,080	0.000	-0.751	1.984
Mobile subscription rate	33,080	83.531	88.984	47.109
Additional tax variables				
Host-country EATR	33,080	0.081	0.096	0.035
International EATR	33,080	0.076	0.040	0.078
EATR without tax incentives	33,080	0.284	0.275	0.048

Source: Author's estimate

Presentation Outline

1. Introduction

2. Empirical Strategy

3. Findings

Baseline result: Taxes are important

VARIABLES	(1) Tax only	(2) (1)+proximity	(3) (2)+economic factors	(4) Baseline (3)+Infrastructure
Bilateral EATR	-2.101***	-2.028***	-5.166***	-5.331***
	(0.332)	(0.347)	(0.562)	(0.563)
Distance (L)		-0.709***	-0.688***	-0.701***
		(0.032)	(0.038)	(0.038)
Colonial		0.144**	0.199***	0.229***
		(0.062)	(0.063)	(0.064)
Real GDP (L)			0.305***	0.350***
			(0.027)	(0.029)
Governance			0.206***	0.159***
			(0.005)	(0.007)
Mobile Subscrip-tion				0.009***
				(0.001)
Observations	33,080	33,080	33,080	33,080
Marginal effects of the tax variable				
Bilateral EATR	-0.336	-0.324	-0.827	-0.853

The impact of tax is significant throughout the specifications

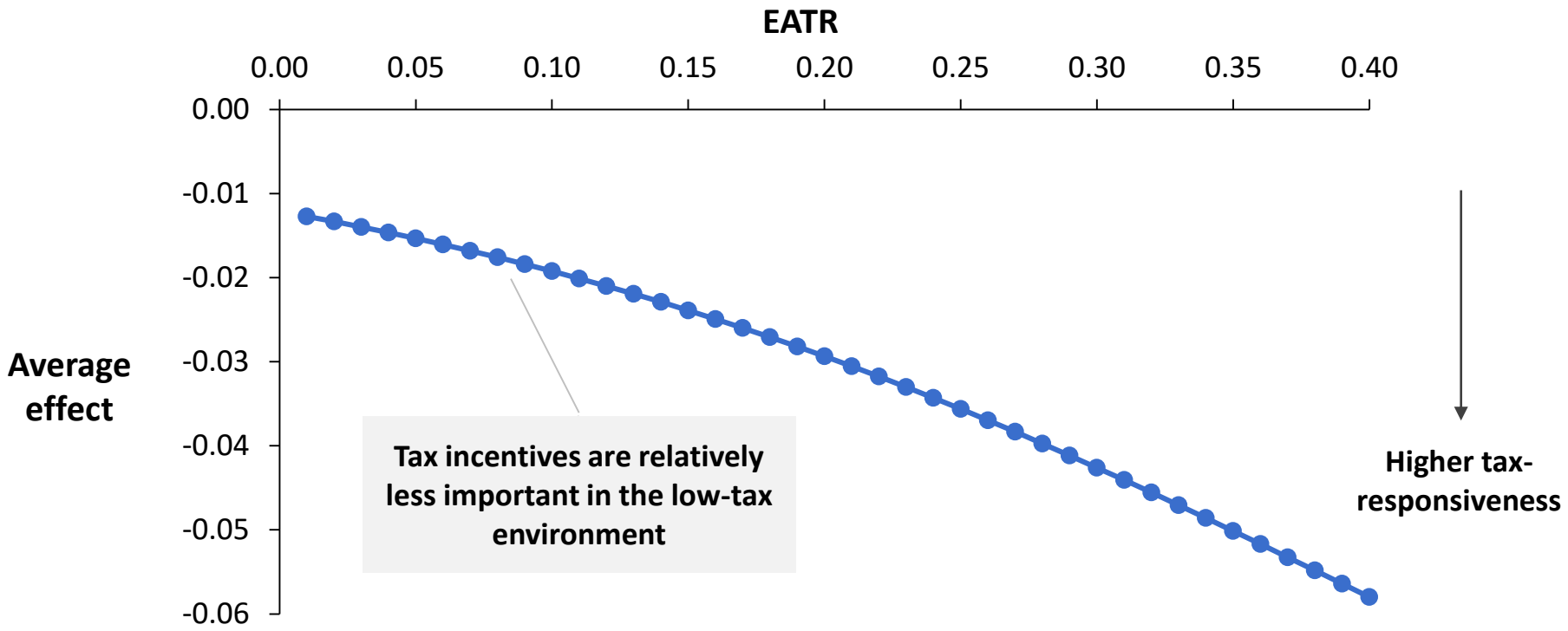
Its economic significance is slightly lower than that for Europe

Notes: ***, ** and * = Significantly different from zero at the 1%, 5% and 10% level, respectively. The marginal effects are evaluated at the means of all control variables.

Examine how the effects of tax policy changes may vary with the existing level of tax rate

Distribution of the average effect of an SD increase in the bilateral EATR

Larger effect (in absolute value) indicates higher tax responsiveness



At very low EATR, extra tax incentives may be less effective

Source: Author's estimates

Robustness tests

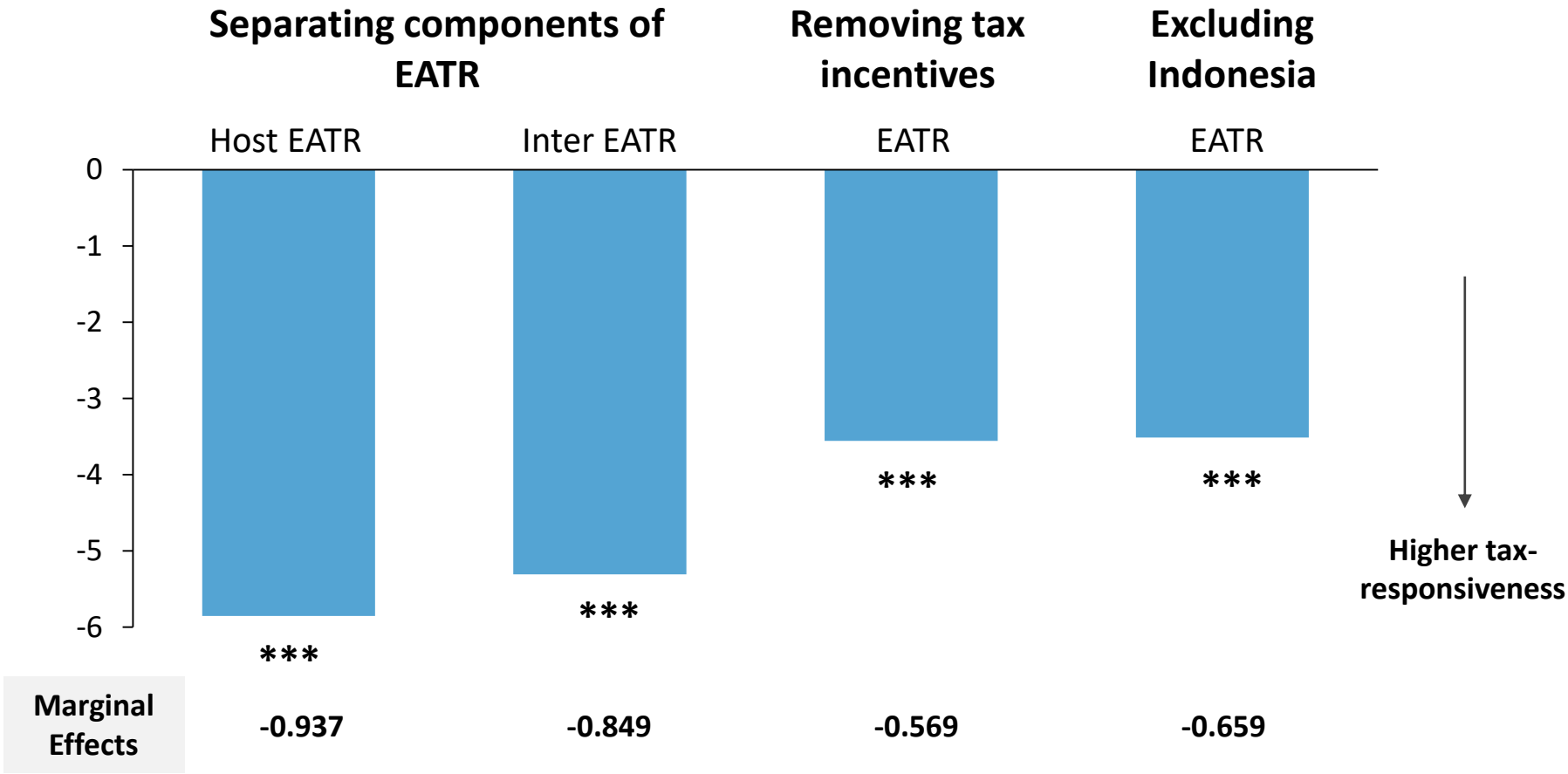
- Perform EBA: Estimate 4,096 regressions to test sensitivity regarding the choice of the control variables in the baseline specification
  - Essentially consider the distribution of estimates on the basis of all possible combinations of other control variable

Variable	Number of regressions	Avg. beta	Avg. SE	% Sig	CDF	95% CI	
						Lower Bound	Upper Bound
EATR	4,096	-6.077	0.831	95.5	0.963	-5.999	-6.155

This provides a strong robustness check of the estimate of the EATR variable to the selection of controls.

Robustness tests

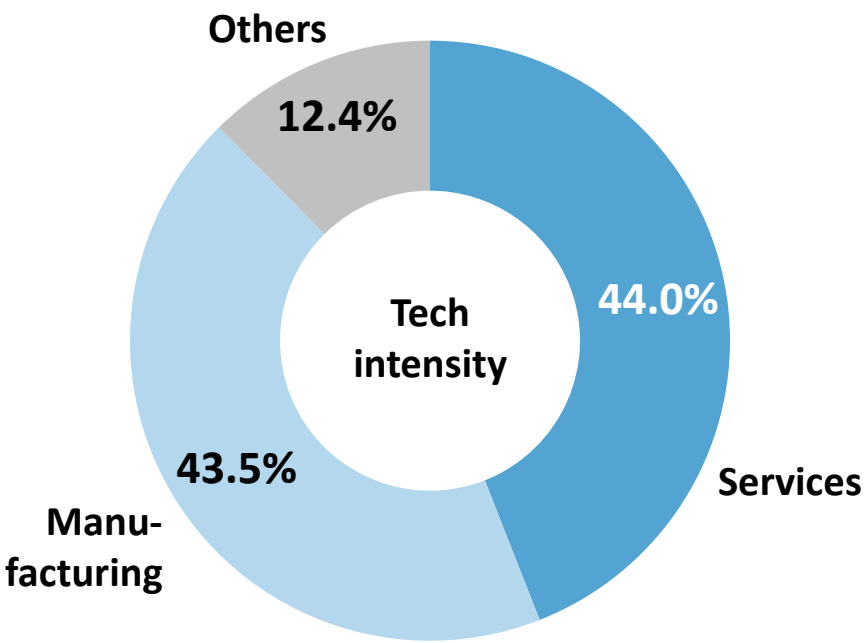
EATR coefficients and statistical significance



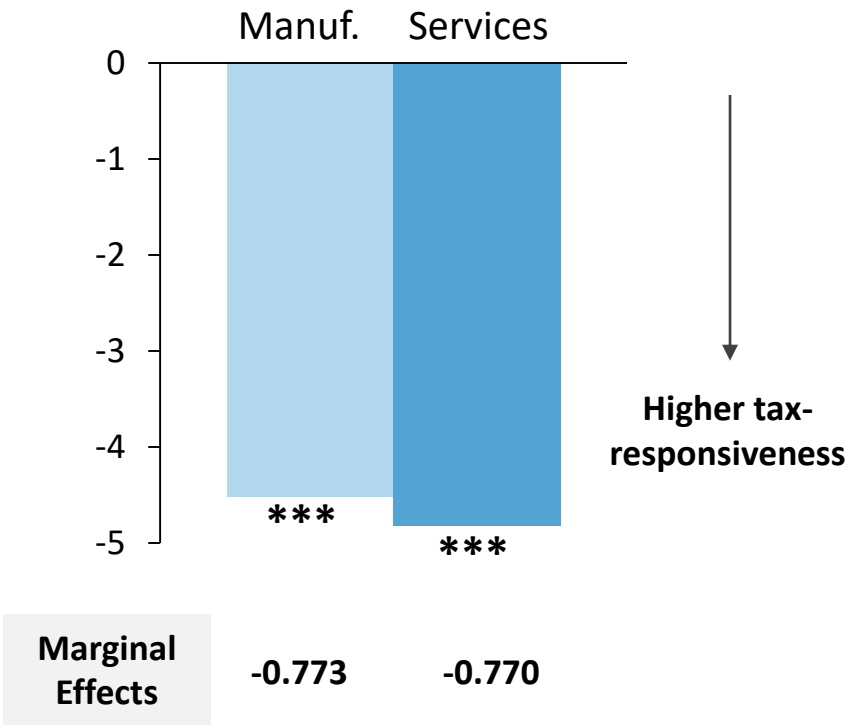
Notes: ***, ** and * = Significantly different from zero at the 1%, 5% and 10% level, respectively. The marginal effects are evaluated at the means of all control variables.

Heterogeneity analysis: Sector

Breakdown of firms



EATR coefficients and statistical significance

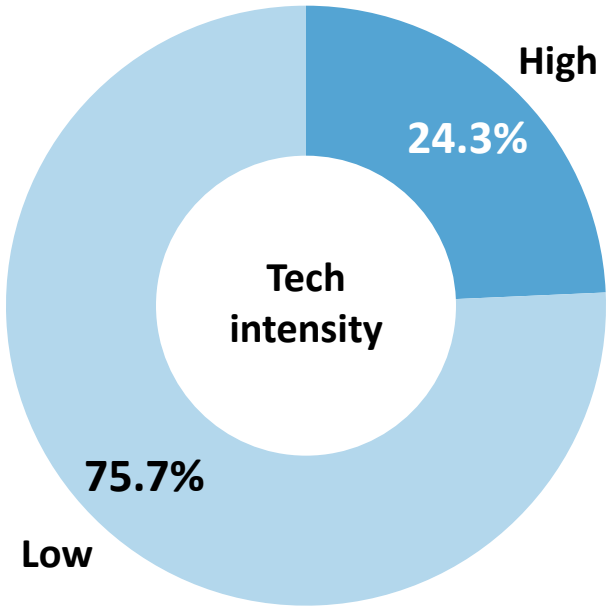


Similar tax responsiveness between manufacturing and services MNEs

Notes: ***, ** and * = Significantly different from zero at the 1%, 5% and 10% level, respectively. The marginal effects are evaluated at the means of all control variables.

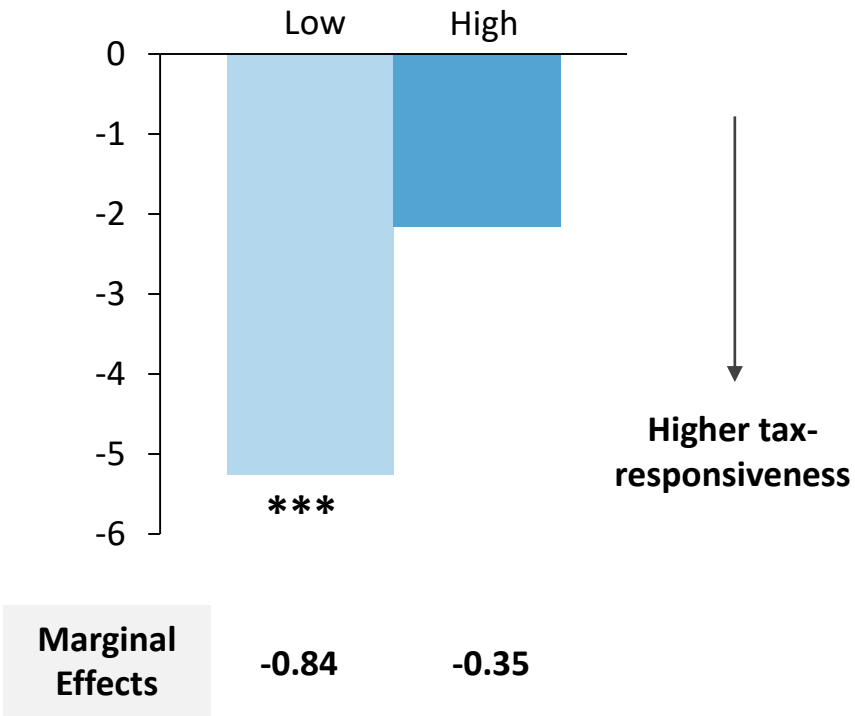
Heterogeneity analysis: Tech intensity

Breakdown of firms



Source: Author's estimate

EATR coefficients and statistical significance

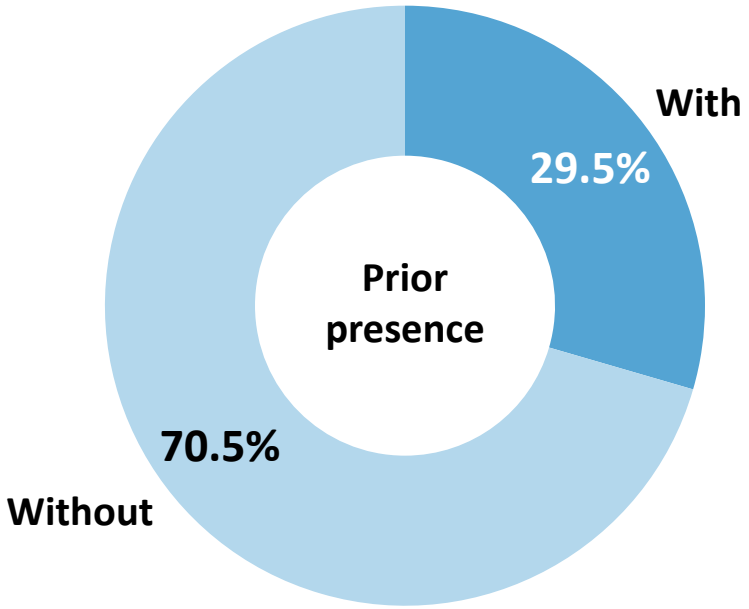


Tax incentives may not be effective for attracting high-tech firms

Note: ***, ** and * = Significantly different from zero at the 1%, 5% and 10% level, respectively. The marginal effects are evaluated at the means of all control variables.

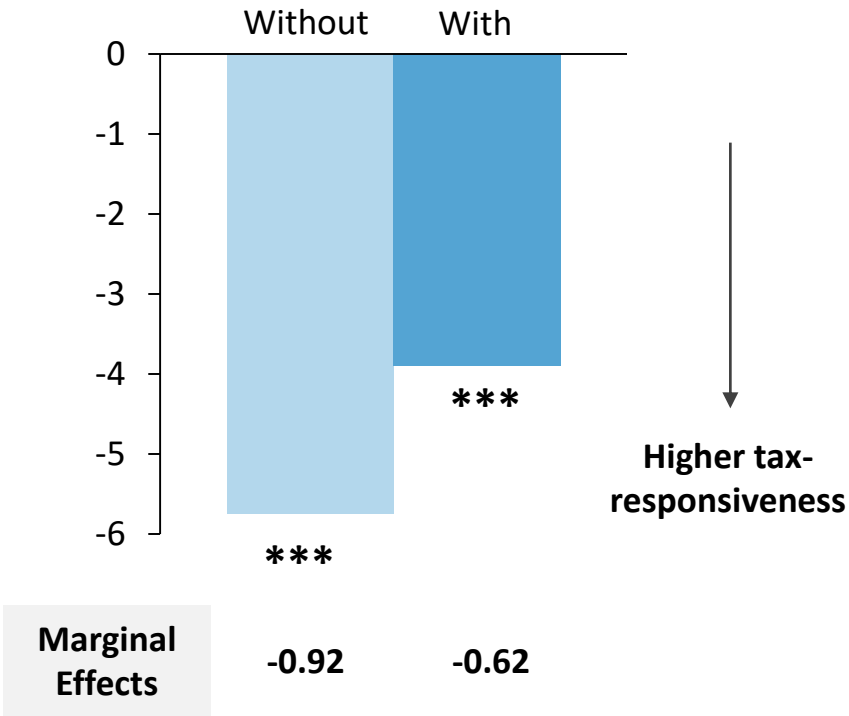
Heterogeneity analysis: Prior presence

Breakdown of firms



Source: Author's estimate

EATR coefficients and statistical significance

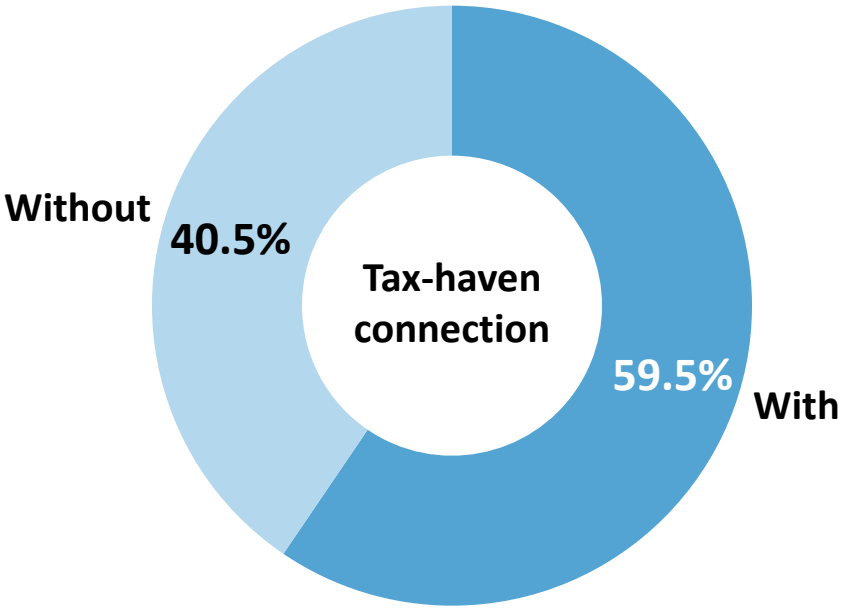


Significantly smaller role of taxation for return investors

Note: ***, ** and * = Significantly different from zero at the 1%, 5% and 10% level, respectively. The marginal effects are evaluated at the means of all control variables.

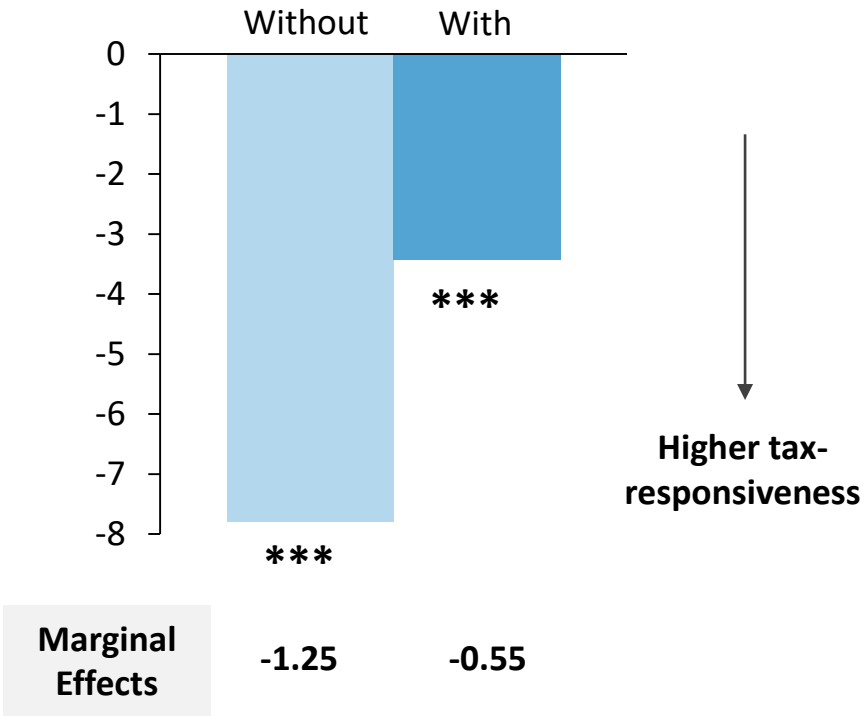
Heterogeneity analysis: Tax-haven connection

Breakdown of firms



Source: Author's estimate

EATR coefficients and statistical significance



Tax havens blunt the tax responsiveness of MNEs

Note: ***, ** and * = Significantly different from zero at the 1%, 5% and 10% level, respectively. The marginal effects are evaluated at the means of all control variables.

Key Takeaways

1. Tax is important but its importance is smaller in the low-tax environment

2. Substantially smaller tax responsiveness for high-tech firms and return investors

3. Tax havens have blunted the importance of tax incentives

End of Document